Learning Objective 9.5: I can determine the reflection and transmission of UPW at a boundary for Transverse Electric (TE) and Transverse Magnetic (TM) waves, using boundary conditions.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate). In every problem the boundary is the plane $z = 0$, region 1 is $z < 0$, the plane of incidence is the xz plane, and both materials are lossless and nonmagnetic, so $\eta = \eta_0/n$ with $\eta_0 = 376.7\ \Omega$.

1. **Phase matching.** A 10 GHz plane wave in air strikes a dielectric with $\epsilon_r = 4$ at $\theta_i = 60°$.

   (a) The tangential electric field at $z = 0$ must satisfy $E_0 e^{-j\beta_1 x \sin\theta_i} + E_0^r e^{-j\beta_1 x \sin\theta_r} = E_0^t e^{-j\beta_2 x \sin\theta_t}$ for every x. Explain why this requires the three exponents to be equal, and derive the law of reflection and Snell's law from it.

   The equation must hold at every point of the boundary. Exponentials in x with different rates drift in and out of step as x changes, so a sum of them cannot equal a third at every x unless all three vary at the same rate. The incident term is not zero, so

   $$\beta_1 \sin\theta_i = \beta_1 \sin\theta_r = \beta_2 \sin\theta_t$$

   The first equality gives $\theta_r = \theta_i$. With $\beta = (\omega/c)n$ and $n = \sqrt{\mu_r \epsilon_r}$, the second gives $n_1 \sin\theta_i = n_2 \sin\theta_t$.

   (b) Find the transmitted angle.

   $n_2 = \sqrt{4} = 2$, so

   $$\sin\theta_t = \frac{\sin 60°}{2} = 0.433 \implies \theta_t = 25.7°$$

   θ_t = $25.7°$

   (c) Find the wavelength in each region, and the spacing along the boundary between points where each wave has the same phase. Show that the two spacings agree.

   spacing = 3.46 cm

   $$\lambda_1 = \frac{c}{f} = 3.00\text{ cm}, \qquad \lambda_2 = \frac{\lambda_1}{2} = 1.50\text{ cm}$$

   $$\frac{\lambda_1}{\sin\theta_i} = \frac{3.00}{0.866} = 3.46\text{ cm}$$

   $$\frac{\lambda_2}{\sin\theta_t} = \frac{1.50}{0.433} = 3.46\text{ cm}$$

2. **The TE coefficients.** The wave of Question 1 is TE, with $\tilde{\mathbf{E}}^i = \hat{\mathbf{y}}\, E_0\, e^{-j\beta_1(x\sin\theta_i + z\cos\theta_i)}$.

(a) Write the reflected and transmitted electric and magnetic fields, using $\tilde{\mathbf{H}} = \hat{\mathbf{k}} \times \tilde{\mathbf{E}}/\eta$. Apply the two tangential boundary conditions and derive Γ_{TE} and τ_{TE}.

$$\tilde{\mathbf{H}}^i = \frac{E_0}{\eta_1}\,(-\hat{\mathbf{x}}\cos\theta_i + \hat{\mathbf{z}}\sin\theta_i)\, e^{-j\beta_1(x\sin\theta_i + z\cos\theta_i)}$$
$$\tilde{\mathbf{E}}^r = \hat{\mathbf{y}}\,\Gamma_{\text{TE}} E_0\, e^{-j\beta_1(x\sin\theta_i - z\cos\theta_i)}$$
$$\tilde{\mathbf{H}}^r = \frac{\Gamma_{\text{TE}} E_0}{\eta_1}\,(\hat{\mathbf{x}}\cos\theta_i + \hat{\mathbf{z}}\sin\theta_i)\, e^{-j\beta_1(x\sin\theta_i - z\cos\theta_i)}$$
$$\tilde{\mathbf{E}}^t = \hat{\mathbf{y}}\,\tau_{\text{TE}} E_0\, e^{-j\beta_2(x\sin\theta_t + z\cos\theta_t)}$$
$$\tilde{\mathbf{H}}^t = \frac{\tau_{\text{TE}} E_0}{\eta_2}\,(-\hat{\mathbf{x}}\cos\theta_t + \hat{\mathbf{z}}\sin\theta_t)\, e^{-j\beta_2(x\sin\theta_t + z\cos\theta_t)}$$

At $z = 0$, phase matching cancels the exponentials. Tangential E_y and H_x:

$$1 + \Gamma_{\text{TE}} = \tau_{\text{TE}}$$
$$\frac{\cos\theta_i}{\eta_1}(1 - \Gamma_{\text{TE}}) = \frac{\cos\theta_t}{\eta_2}\tau_{\text{TE}}$$
$$\eta_2\cos\theta_i\,(1 - \Gamma_{\text{TE}}) = \eta_1\cos\theta_t\,(1 + \Gamma_{\text{TE}})$$
$$\Gamma_{\text{TE}} = \frac{\eta_2\cos\theta_i - \eta_1\cos\theta_t}{\eta_2\cos\theta_i + \eta_1\cos\theta_t}$$
$$\tau_{\text{TE}} = 1 + \Gamma_{\text{TE}} = \frac{2\eta_2\cos\theta_i}{\eta_2\cos\theta_i + \eta_1\cos\theta_t}$$

(b) Evaluate Γ_{TE} and τ_{TE}.

$\eta_2 = 376.7/2 = 188.4\,\Omega$.

$$\Gamma_{\text{TE}} = \frac{188.4\times 0.500 - 376.7\times 0.901}{188.4\times 0.500 + 376.7\times 0.901} = \frac{94.2 - 339.6}{94.2 + 339.6} = -0.566$$
$$\tau_{\text{TE}} = 1 + \Gamma_{\text{TE}} = 0.434$$

Γ_{TE} = -0.566

τ_{TE} = 0.434

(c) Find the fraction of the incident power that reflects. Check that the transmitted fraction, $|\tau_{\text{TE}}|^2(\eta_1/\eta_2)(\cos\theta_t/\cos\theta_i)$, accounts for the rest.

reflected = 32.0%

$$|\Gamma_{\text{TE}}|^2 = 0.320$$
$$0.434^2 \times 2 \times \frac{0.901}{0.500} = 0.680 = 1 - 0.320$$

3. **Limits and a second boundary.**

(a) Show that Γ_{TE} reduces to Lesson 38's normal-incidence Γ when $\theta_i = 0$, and find its limit as $\theta_i \to 90^\circ$.

At $\theta_i = 0$, Snell's law gives $\theta_t = 0$, both cosines are one, and $\Gamma_{\text{TE}} = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$. As $\theta_i \to 90^\circ$, $\cos\theta_i \to 0$ while $\cos\theta_t$ stays positive, so $\Gamma_{\text{TE}} \to -\eta_1 \cos\theta_t / (\eta_1 \cos\theta_t) = -1$.

(b) A TE wave in glass, $n_1 = 1.50$, meets water, $n_2 = 1.33$, at $\theta_i = 20^\circ$. Find θ_t, Γ_{TE}, and τ_{TE}.

$$\sin\theta_t = \frac{1.50}{1.33}\sin 20^\circ = 0.386 \implies \theta_t = 22.7^\circ$$

$$\eta_1 = 251.2\,\Omega, \qquad \eta_2 = 283.3\,\Omega$$

$$\Gamma_{\text{TE}} = \frac{283.3 \times 0.940 - 251.2 \times 0.923}{283.3 \times 0.940 + 251.2 \times 0.923} = 0.069$$

$$\tau_{\text{TE}} = 1.069$$

θ_t = 22.7°

Γ_{TE} = 0.069

(c) Explain how τ_{TE} can be greater than one without the transmitted wave carrying more power than the incident wave.

τ_{TE} is a ratio of electric fields. Water has the larger intrinsic impedance, and a field of a given amplitude carries less power in a material with a larger η. The reflected power is $0.069^2 = 0.5\%$, and the transmitted wave carries the other 99.5%.

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